

DEEPAK KUMAR

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PROFESSIONAL SUMMARY

Electrical & Electronics Engineering undergraduate trained in IoT, embedded systems, Arduino, ESP32 and solar PV technology. Hands-on with sensor interfacing, hardware integration, circuit design, C and Embedded C programming, MATLAB and Tinkercad simulation, and GitHub-based version control, plus solar PV design, installation, operation and maintenance. Seeking an internship or Graduate Engineer Trainee (GET) role in IoT, embedded systems, industrial automation or electrical systems.

TECHNICAL SKILLS

Hardware & Microcontrollers: Arduino UNO, ESP32, ESP32-CAM, ESP8266

Programming & Tools: C, Embedded C, MATLAB, Tinkercad, GitHub, AutoCAD Electrical (Learning)

Communication & Connectivity: Bluetooth, Wi-Fi, RFID

Sensors & Modules: IR Sensor, Ultrasonic Sensor, Motion Sensor, GPS, RTC DS3231, 7-Segment Display, Fingerprint Sensor

Electrical & Energy: Electrical Circuit Design, Solar PV Systems Design & Installation, Sensor Interfacing, Hardware Integration, System Debugging

INTERNSHIP EXPERIENCE

IoT with Project — Jharkhand Government Tool Room, Ranchi | 27 May 2025 – 26 June 2025

- Interfaced and tested IR, ultrasonic and motion sensors with 7-segment displays, Bluetooth modules and DC motor circuits; systematic circuit checks cut manual debugging time.
- Built RFID access control with event logging, fingerprint voting logic, GPS tracking and RTC (DS3231) time display across 4 independent mini-modules.
- Configured ESP8266 Wi-Fi and ESP32-CAM for live wireless video streaming, checking range and stability at several test locations.

Web Development Internship — VaultofCodes.in | July 2024 · 1 Month

- Covered HTML, CSS and front-end structuring fundamentals, enough to work on personal and portfolio sites.

Solar PV Skills Training — Tata Power Skill Development Institute (TPSDI) | TVAS Certified Solar PV Skills Professional

- Training covered Solar PV Fundamentals, Off-Grid, On-Grid and Hybrid systems.
- Studied Solar PV design, installation, operation, maintenance and commissioning.
- Built a practical understanding of solar energy systems and renewable energy applications.
- Completed the Solar PV Professional Advantage Course.

TECHNICAL PROJECTS

Smart IoT Monitoring System using ESP32-CAM | Academic / Practical Project

- Designed an end-to-end IoT monitoring system on the ESP32-CAM module, integrating multiple sensors for real-time data acquisition over Wi-Fi.
- Ran hardware integration, functional testing and debugging cycles to improve reliability and reduce false sensor triggers.

Arduino & IoT-Based Embedded Applications | Practical Project Set

- Built 6+ standalone embedded applications: LED control, 7-segment display, IR remote input, ultrasonic distance measurement and motion-triggered detection.
- Developed a Bluetooth-controlled car with obstacle detection and automated motor control response.
- Implemented RFID access control with alert logging, fingerprint voting, GPS tracking and RTC time display modules.

EDUCATION

B.Tech in Electrical & Electronics Engineering (EEE) | 2023 – 2027

Cambridge Institute of Technology, Ranchi (Jharkhand University of Technology) | CGPA: 7.27 / 10 (through 5th Semester)

Intermediate (Science) | 2023

Rajiv Gandhi Inter College, Pindra Tapin, Ramgarh | 85.8% — First Division

Matriculation (Secondary) | 2021

Meridian Academy, Jhumri Telaiya, Koderma | Status: Pass

CERTIFICATIONS & TRAININGS

- Internship on IoT with Project — Jharkhand Government Tool Room, Ranchi
- Skill Development Workshop on IoT & Robotics — Jharkhand University of Technology / Nextlayer LLP
- Workshop on Emerging Control & Digital Technologies — BIT Mesra